TermTrivia sequence (Phase 3): two clients, one game, a stalled client in another game

Host client A | **Player client B** | **Handler threads A, B** | **Game 1 lane + timer** | **Game 2 (other game)**

1. Both clients connect; each gets its own handler thread, so neither waits for the other

TCP connect (handler thread 1 starts)

TCP connect (handler thread 2 starts, at the same time)

CREATE_GAME Robert 3

new game 1: lane held until the 200 is written

200 OK / 7QXA / token

JOIN_GAME 7QXA Matthew

200 OK / 2 players / token

2. Start: the reply is always written before the first push, to every player

START_ROUND

lane 1 held: reply, then question pushes

200 OK

PUSH QUESTION 1

PUSH QUESTION 1

3. Both answer at the same moment: each answer counts once

SUBMIT_ANSWER A

SUBMIT_ANSWER B

handlers queue on lane 1, one at a time

200 OK

200 OK

PUSH ROUND_RESULT 1 (all answered)

PUSH ROUND_RESULT 1

4. Another game is unaffected by a stalled reader in game 1

client B stops reading; writes to B time out (200 ms)

LEADERBOARD (game 2 client)

200 OK (under 500 ms: game 2 has its own lane)

5. Unexpected disconnect and reconnect

B is killed (no QUIT)

B kept as DISCONNECTED_PENDING for 60 s; game continues

RECONNECT Matthew 7QXA token (new connection)

200 OK / score / question number

PUSH QUESTION n (if a round is open)

6. Game ends: every connected player gets the push, then the sockets close

timer: last round closed, final intermission over

PUSH GAME_END COMPLETE / FINAL RESULTS

PUSH GAME_END COMPLETE / FINAL RESULTS

server closes both sockets; game 1 and its lane are removed

HOST_EXPIRED: if the host never returns in the lobby, players get GAME_END HOST_EXPIRED instead

Legend: request | reply (one per request) | push (starts with PUSH)